10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 1.2 — Algebraic Operations

Unit 1 | Solved Exercise | New Book (PECTAA)

Exercise 1.2 (Solved)

1. Simplify and write in the form a + bi:

(i) (2+5i) + (3−2i)

(ii) (16−3i) + (9+2i)

(iii) (9−2i) − (7+3i)

(iv) (11+9i) − (9−7i)

(v) (3+4i)(2−3i)

(vi) (5−2i)(3−4i)

(vii) (3−5i) ÷ (2−4i)

(viii) (5+2i) ÷ (6−3i)

(i) (2 + 5i) + (3 − 2i)

= (2 + 3) + (5 − 2)i

Answer: 5 + 3i

(ii) (16 − 3i) + (9 + 2i)

= (16 + 9) + (−3 + 2)i

Answer: 25 − i

(iii) (9 − 2i) − (7 + 3i)

= 9 − 2i − 7 − 3i

= (9 − 7) + (−2 − 3)i

Answer: 2 − 5i

(iv) (11 + 9i) − (9 − 7i)

= 11 + 9i − 9 + 7i

= (11 − 9) + (9 + 7)i

Answer: 2 + 16i

(v) (3 + 4i)(2 − 3i)

= 3(2 − 3i) + 4i(2 − 3i)

= 6 − 9i + 8i − 12i^2

= 6 − i − 12(−1) ∵ $i^2 = -1$

= 6 − i + 12

Answer: 18 − i

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(vi) (5 − 2i)(3 − 4i)

$= 15 - 20i - 6i + 8i^2$

$= 15 - 26i + 8(-1)$

$= 15 - 26i - 8$

Answer: 7 − 26i

(vii) (3 − 5i) ÷ (2 − 4i)

$= \frac{3 - 5i}{2 - 4i}$

Multiply the numerator and the denominator by the conjugate of the denominator, that is 2 + 4i.

$= \frac{3 - 5i}{2 - 4i} \times \frac{2 + 4i}{2 + 4i}$

$= \frac{6 + 12i - 10i - 20i^2}{(2)^2 - (4i)^2}$

$= \frac{6 + 2i + 20}{4 + 16} \quad \because i^2 = -1$

$= \frac{26 + 2i}{20}$

$= \frac{26}{20} + \frac{2}{20}\, i$

Answer: 13/10 + 1/10 i

(viii) (5 + 2i) ÷ (6 − 3i)

$= \frac{5 + 2i}{6 - 3i} \times \frac{6 + 3i}{6 + 3i}$

$= \frac{30 + 15i + 12i + 6i^2}{(6)^2 - (3i)^2}$

$= \frac{30 + 27i - 6}{36 + 9} \quad \because i^2 = -1$

$= \frac{24 + 27i}{45}$

$= \frac{24}{45} + \frac{27}{45}\, i$

Answer: 8/15 + 3/5 i

2. Write additive inverse for each complex number:

(i) 3 + 2i (ii) 4 − 3i (iii) 5 − 7i

(iv) $-\frac{2}{3} + \frac{5}{4}\, i$

The additive inverse of z is −z, because z + (−z) = 0.

(i) 3 + 2i

Additive inverse = −(3 + 2i)

= −3 − 2i

Answer: −3 − 2i

(ii) 4 − 3i

Additive inverse = −(4 − 3i)

= −4 + 3i

Answer: −4 + 3i

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(iii) 5 − 7i

Additive inverse = −(5 − 7i)

= −5 + 7i

Answer: −5 + 7i

(iv) −2/3 + 5/4 i

Additive inverse = $-(-\frac{2}{3} + \frac{5}{4}\ i)$

= $\frac{2}{3} - \frac{5}{4}\ i$

Answer: 2/3 − 5/4 i

3. Find multiplicative inverse for each complex number:

(i) 4 + 5i (ii) 6 + 2i (iii) 7 − 3i

(iv) √5 − 4i

The multiplicative inverse of z is $z^{-1} = 1/z$, which is found by multiplying the numerator and the denominator by the conjugate of z.

(i) 4 + 5i

$z^{-1} = \frac{1}{4 + 5i}$

$= \frac{1}{4 + 5i} \times \frac{4 - 5i}{4 - 5i}$

$= \frac{4 - 5i}{(4)^2 - (5i)^2} = \frac{4 - 5i}{16 + 25}$

$= \frac{4 - 5i}{41}$

Answer: 4/41 − 5/41 i

(ii) 6 + 2i

$z^{-1} = \frac{1}{6 + 2i} \times \frac{6 - 2i}{6 - 2i}$

$= \frac{6 - 2i}{36 - 4i^2} = \frac{6 - 2i}{36 + 4}$

$= \frac{6 - 2i}{40}$

$= \frac{6}{40} - \frac{2}{40}\ i$

$= \frac{3}{20} - \frac{1}{20}\ i$

Answer: 3/20 − 1/20 i

(iii) 7 − 3i

$z^{-1} = \frac{1}{7 - 3i} \times \frac{7 + 3i}{7 + 3i}$

$= \frac{7 + 3i}{49 - 9i^2} = \frac{7 + 3i}{49 + 9}$

$= \frac{7 + 3i}{58}$

Answer: 7/58 + 3/58 i

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(iv) $\sqrt{5} - 4i$

$$z^{-1} = \frac{1}{\sqrt{5} - 4i} \times \frac{\sqrt{5} + 4i}{\sqrt{5} + 4i}$$
$$= \frac{\sqrt{5} + 4i}{(\sqrt{5})^2 - (4i)^2} = \frac{\sqrt{5} + 4i}{5 + 16}$$
$$= \frac{\sqrt{5} + 4i}{21}$$

Answer: √5/21 + 4/21 i

4. If $z_1 = 2 + 5i$, $z_2 = 1 - 3i$ and $z_3 = 2 + i$, then verify that:

(i) $z_1 + z_2 = z_2 + z_1$ (ii) $z_1z_2 = z_2z_1$

(iii) $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ (iv) $(z_1z_2)z_3 = z_1(z_2z_3)$

(v) $z_1 + (-z_1) = (-z_1) + z_1 = 0$

(i) $z_1 + z_2 = z_2 + z_1$ (commutative law)

L.H.S. $= z_1 + z_2$

$= (2 + 5i) + (1 - 3i)$

$= (2 + 1) + (5 - 3)i$

$= 3 + 2i$

R.H.S. $= z_2 + z_1$

$= (1 - 3i) + (2 + 5i)$

$= (1 + 2) + (-3 + 5)i$

$= 3 + 2i$

Answer: L.H.S. = R.H.S., hence verified

(ii) $z_1z_2 = z_2z_1$ (commutative law)

$z_1z_2 = (2 + 5i)(1 - 3i)$

$= 2 - 6i + 5i - 15i^2$

$= 2 - i + 15$

$= 17 - i$

$z_2z_1 = (1 - 3i)(2 + 5i)$

$= 2 + 5i - 6i - 15i^2$

$= 2 - i + 15$

$= 17 - i$

Answer: L.H.S. = R.H.S., hence verified

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(iii) $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ (associative law)

$z_1 + z_2 = 3 + 2i$ (from part i)

L.H.S. $= (3 + 2i) + (2 + i)$

$= (3 + 2) + (2 + 1)i$

$= 5 + 3i$

$z_2 + z_3 = (1 - 3i) + (2 + i)$

$= (1 + 2) + (-3 + 1)i$

$= 3 - 2i$

R.H.S. $= z_1 + (3 - 2i)$

$= (2 + 5i) + (3 - 2i)$

$= (2 + 3) + (5 - 2)i$

$= 5 + 3i$

Answer: L.H.S. = R.H.S., hence verified

(iv) $(z_1z_2)z_3 = z_1(z_2z_3)$ (associative law)

$z_1z_2 = 17 - i$ (from part ii)

L.H.S. $= (17 - i)(2 + i)$

$= 34 + 17i - 2i - i^2$

$= 34 + 15i + 1$

$= 35 + 15i$

$z_2z_3 = (1 - 3i)(2 + i)$

$= 2 + i - 6i - 3i^2$

$= 2 - 5i + 3$

$= 5 - 5i$

R.H.S. $= z_1(5 - 5i)$

$= (2 + 5i)(5 - 5i)$

$= 10 - 10i + 25i - 25i^2$

$= 10 + 15i + 25$

$= 35 + 15i$

Answer: L.H.S. = R.H.S., hence verified

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(v) $z_1 + (-z_1) = (-z_1) + z_1 = 0$

$z_1 = 2 + 5i$

$-z_1 = -2 - 5i$

$z_1 + (-z_1) = (2 - 2) + (5 - 5)i$

$= 0 + 0i$

$= 0$

$(-z_1) + z_1 = (-2 + 2) + (-5 + 5)i$

$= 0 + 0i$

$= 0$

Answer: $z_1 + (-z_1) = (-z_1) + z_1 = 0$, hence verified

5. If $(1 + i)^2 \div (2 - i) = x + iy$, then find the values of x and y.

$(1 + i)^2 = 1 + 2i + i^2$

$= 1 + 2i - 1$

$= 2i$

$\therefore \quad \frac{2i}{2 - i} = \frac{2i}{2 - i} \times \frac{2 + i}{2 + i}$

$= \frac{4i + 2i^2}{(2)^2 - (i)^2} = \frac{4i - 2}{4 + 1}$

$= \frac{-2 + 4i}{5}$

$= -\frac{2}{5} + \frac{4}{5}\, i$

Comparing with $x + iy$: $x = -\frac{2}{5}$, $y = \frac{4}{5}$

Answer: x = −2/5, y = 4/5

6. If $(2x + iy)(1 - i) = 4 + 2i$, then find the values of x and y.

$(2x + iy)(1 - i) = 2x - 2xi + iy - i^2y$

$= (2x + y) + (-2x + y)i$

$\therefore \; (2x + y) + (-2x + y)i = 4 + 2i$

Real parts: $2x + y = 4$ ... (1)

Imaginary parts: $-2x + y = 2$ ... (2)

Adding (1) and (2): $2y = 6 \implies y = 3$

From (1): $2x + 3 = 4$

$2x = 1$

$x = \frac{1}{2}$

Answer: x = 1/2, y = 3

7. Find the values of a and b, if $(a + bi)(1 + 3i) = -8 + 11i$.

$(a + bi)(1 + 3i) = a + 3ai + bi + 3bi^2$

$= (a - 3b) + (3a + b)i$

$\therefore \; (a - 3b) + (3a + b)i = -8 + 11i$

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Real parts: $a - 3b = -8$ ... (1)

Imaginary parts: $3a + b = 11$ ... (2)

Multiply (2) by 3: $9a + 3b = 33$ ... (3)

Add (1) and (3): $10a = 25 \implies a = \frac{5}{2}$

From (2): $3(\frac{5}{2}) + b = 11$

$\frac{15}{2} + b = 11$

$b = \frac{7}{2}$

Answer: a = 5/2, b = 7/2

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